

eda_life_style_data

November 5, 2025

1 Life_Style_Data.csv

1.1 Chargement et inspection initiale

1.1.1 Objectif

Vérifier que le fichier `Life_Style_Data.csv` est correctement chargé et obtenir une vue d'ensemble :
- dimensions du jeu de données (nombre de lignes et de colonnes), - typage des variables (numériques, catégorielles, booléennes), - détection des valeurs manquantes et doublons, - cohérence générale du corpus avant analyse statistique.

1.1.2 Interprétation attendue

- Le dataset devrait comporter une cinquantaine de colonnes, combinant données **physiques, comportementales et nutritionnelles**.
- Les types doivent être correctement inférés : numériques (`Age`, `Weight`, `Calories_Burned`), catégorielles (`Gender`, `Workout_Type`), booléennes (`is_healthy`).
- On identifiera les **colonnes à forte proportion de valeurs manquantes** afin de décider ultérieurement d'un traitement (imputation, suppression, ou encodage spécial).
- La détection de **doublons** permettra d'évaluer la fiabilité du corpus et la diversité des observations.
- Enfin, l'inspection des noms de colonnes garantira la cohérence du dictionnaire des données avant les analyses statistiques.

1.1.3 Importation des librairies essentielles

```
[1]: import pandas as pd  
import numpy as np
```

1.1.4 Configuration d'affichage pour plus de lisibilité

```
[2]: pd.set_option('display.max_columns', None)  
pd.set_option('display.width', 120)
```

1.1.5 Chargement du dataset

```
[3]: # Construction du chemin relatif
file_path = "../../../datasets/Life_Style_Data.csv"

# Lecture du fichier CSV
df = pd.read_csv(file_path)

# Informations de confirmation
print("Dataset chargé avec succès !")
print(f"Dimensions : {df.shape[0]} lignes et {df.shape[1]} colonnes\n")
```

Dataset chargé avec succès !

Dimensions : 20000 lignes et 54 colonnes

1.1.6 Aperçu général des premières lignes

```
[4]: display(df.head())
```

	Age	Gender	Weight (kg)	Height (m)	Max_BPM	Avg_BPM	Resting_BPM	
	Session_Duration (hours)		Calories_Burned \					
0	34.91	Male	65.27	1.62	188.58	157.65	69.05	
		1.00		1080.90				
1	23.37	Female	56.41	1.55	179.43	131.75	73.18	
		1.37		1809.91				
2	33.20	Female	58.98	1.67	175.04	123.95	54.96	
		0.91		802.26				
3	38.69	Female	93.78	1.70	191.21	155.10	50.07	
		1.10		1450.79				
4	45.09	Male	52.42	1.88	193.58	152.88	70.84	
		1.08		1166.40				
	Workout_Type		Fat_Percentage		Water_Intake (liters)		Workout_Frequency (days/	
	week)		Experience_Level		BMI \			
0	Strength		26.800377		1.50			
	3.99		2.01 24.87					
1	HIIT		27.655021		1.90			
	4.00		2.01 23.48					
2	Cardio		24.320821		1.88			
	2.99		1.02 21.15					
3	HIIT		32.813572		2.50			
	3.99		1.99 32.45					
4	Strength		17.307319		2.91			
	4.00		2.00 14.83					
	Daily meals frequency		Physical exercise		Carbs		Proteins	
	meal_name meal_type		diet_type \				Fats	
							Calories	

0		2.99		0.01	267.68	106.05	71.63	1806.0	␣
↪	Other	Lunch	Vegan						
1		3.01		0.97	214.32	85.41	56.97	1577.0	␣
↪	Other	Lunch	Vegetarian						
2		1.99		-0.02	246.04	98.11	65.48	1608.0	␣
↪	Other	Breakfast	Paleo						
3		3.00		0.04	203.22	80.84	54.56	2657.0	␣
↪	Other	Lunch	Paleo						
4		3.00		3.00	332.79	133.05	88.43	1470.0	␣
↪	Other	Breakfast	Vegan						

	sugar_g	sodium_mg	cholesterol_mg	serving_size_g	cooking_method	␣
↪	prep_time_min	cook_time_min	rating	\		
0	31.77	1729.94	285.05	120.47	Grilled	16.
↪24		110.79	1.31			
1	12.34	693.08	300.61	109.15	Fried	16.
↪47		12.01	1.92			
2	42.81	2142.48	215.42	399.43	Boiled	54.
↪35		6.09	4.70			
3	9.34	123.20	9.70	314.31	Fried	27.
↪73		103.72	4.85			
4	23.78	1935.11	116.89	99.22	Baked	34.
↪16		46.55	3.07			

	Name of Exercise	Sets	Reps	␣
↪	Benefit	Burns Calories (per 30 min)	\	
0	Decline Push-ups	4.99	20.91	Improves shoulder health and
↪	posture		342.58	
1	Bear Crawls	4.01	16.15	Strengthens lower
↪	abs		357.16	
2	Dips	5.00	21.90	Builds chest
↪	strength		359.63	
3	Mountain Climbers	4.01	16.92	Improves coordination and cardiovascular
↪	health		351.65	
4	Bicep Curls	4.99	15.01	Targets obliques and improves core
↪	rotation		329.36	

	Target Muscle Group	Equipment Needed	Difficulty Level	Body Part	␣
↪	Type of Muscle	Workout	\		
0	Shoulders, Triceps	Cable Machine	Advanced	Legs	␣
↪	Lats	Dumbbell flyes			
1	Back, Core, Shoulders	Step or Box	Intermediate	Chest	␣
↪	Lats	Lateral raises			
2	Quadriceps, Glutes	Step or Box	Intermediate	Arms	␣
↪	Grip Strength	Standing calf raises			

3	Biceps, Forearms	Parallel Bars or Chair	Advanced	Shoulders	┐
↪	Upper	Incline dumbbell flyes			
4	Chest, Triceps	Wall	Advanced	Abs	┐
↪	Wrist Flexors	Military press			

	BMI_calc	cal_from_macros	pct_carbs	protein_per_kg	pct_HRR	pct_maxHR	┐
↪	cal_balance	lean_mass_kg \					
0	24.870447	2139.59	0.500432	1.624789	0.741237	0.835985	┐
↪	725.10	47.777394					
1	23.479709	1711.65	0.500850	1.514093	0.551247	0.734270	┐
↪	-232.91	40.809803					
2	21.148123	1965.92	0.500610	1.663445	0.574534	0.708124	┐
↪	805.74	44.635580					
3	32.449827	1627.28	0.499533	0.862017	0.744155	0.811150	┐
↪	1206.21	63.007432					
4	14.831372	2659.23	0.500581	2.538153	0.668405	0.789751	┐
↪	303.60	43.347504					

	expected_burn	Burns Calories (per 30 min)_bc	Burns_Calories_Bin
0	685.1600	7.260425e+19	Medium
1	978.6184	1.020506e+20	High
2	654.5266	1.079607e+20	High
3	773.6300	8.987921e+19	High
4	711.4176	5.264685e+19	Low

1.1.7 Informations générales sur le typage et la complétude

```
[5]: df_info = df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20000 entries, 0 to 19999
Data columns (total 54 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Age                                         20000 non-null  float64
1   Gender                                     20000 non-null  object
2   Weight (kg)                               20000 non-null  float64
3   Height (m)                                20000 non-null  float64
4   Max_BPM                                    20000 non-null  float64
5   Avg_BPM                                    20000 non-null  float64
6   Resting_BPM                               20000 non-null  float64
7   Session_Duration (hours)                  20000 non-null  float64
8   Calories_Burned                           20000 non-null  float64
9   Workout_Type                              20000 non-null  object
10  Fat_Percentage                             20000 non-null  float64
11  Water_Intake (liters)                      20000 non-null  float64
12  Workout_Frequency (days/week)             20000 non-null  float64
```

13	Experience_Level	20000	non-null	float64
14	BMI	20000	non-null	float64
15	Daily meals frequency	20000	non-null	float64
16	Physical exercise	20000	non-null	float64
17	Carbs	20000	non-null	float64
18	Proteins	20000	non-null	float64
19	Fats	20000	non-null	float64
20	Calories	20000	non-null	float64
21	meal_name	20000	non-null	object
22	meal_type	20000	non-null	object
23	diet_type	20000	non-null	object
24	sugar_g	20000	non-null	float64
25	sodium_mg	20000	non-null	float64
26	cholesterol_mg	20000	non-null	float64
27	serving_size_g	20000	non-null	float64
28	cooking_method	20000	non-null	object
29	prep_time_min	20000	non-null	float64
30	cook_time_min	20000	non-null	float64
31	rating	20000	non-null	float64
32	Name of Exercise	20000	non-null	object
33	Sets	20000	non-null	float64
34	Reps	20000	non-null	float64
35	Benefit	20000	non-null	object
36	Burns Calories (per 30 min)	20000	non-null	float64
37	Target Muscle Group	20000	non-null	object
38	Equipment Needed	20000	non-null	object
39	Difficulty Level	20000	non-null	object
40	Body Part	20000	non-null	object
41	Type of Muscle	20000	non-null	object
42	Workout	20000	non-null	object
43	BMI_calc	20000	non-null	float64
44	cal_from_macros	20000	non-null	float64
45	pct_carbs	20000	non-null	float64
46	protein_per_kg	20000	non-null	float64
47	pct_HRR	20000	non-null	float64
48	pct_maxHR	20000	non-null	float64
49	cal_balance	20000	non-null	float64
50	lean_mass_kg	20000	non-null	float64
51	expected_burn	20000	non-null	float64
52	Burns Calories (per 30 min)_bc	20000	non-null	float64
53	Burns_Calories_Bin	20000	non-null	object

dtypes: float64(39), object(15)

memory usage: 8.2+ MB

1.1.8 Statistiques descriptives des variables numériques

```
[6]: display(df.describe().T)
```

	count	mean	std	
min	25%	50%		
Age	20000.0	3.885145e+01	1.211458e+01	1.
↳800000e+01	2.817000e+01	3.986500e+01		
Weight (kg)	20000.0	7.389883e+01	2.117301e+01	3.
↳918000e+01	5.816000e+01	7.000000e+01		
Height (m)	20000.0	1.723094e+00	1.270328e-01	1.
↳490000e+00	1.620000e+00	1.710000e+00		
Max_BPM	20000.0	1.798897e+02	1.151081e+01	1.
↳593100e+02	1.700575e+02	1.801400e+02		
Avg_BPM	20000.0	1.437043e+02	1.426769e+01	1.
↳190700e+02	1.312200e+02	1.429900e+02		
Resting_BPM	20000.0	6.219581e+01	7.289351e+00	4.
↳949000e+01	5.596000e+01	6.220000e+01		
Session_Duration (hours)	20000.0	1.259446e+00	3.413362e-01	4.
↳900000e-01	1.050000e+00	1.270000e+00		
Calories_Burned	20000.0	1.280110e+03	5.022290e+02	3.
↳231100e+02	9.108000e+02	1.231450e+03		
Fat_Percentage	20000.0	2.610131e+01	4.996160e+00	1.
↳133313e+01	2.238781e+01	2.582250e+01		
Water_Intake (liters)	20000.0	2.627486e+00	6.047240e-01	1.
↳460000e+00	2.170000e+00	2.610000e+00		
Workout_Frequency (days/week)	20000.0	3.318629e+00	9.111979e-01	1.
↳940000e+00	2.980000e+00	3.010000e+00		
Experience_Level	20000.0	1.809176e+00	7.362036e-01	1.
↳000000e+00	1.010000e+00	1.990000e+00		
BMI	20000.0	2.492168e+01	6.701126e+00	1.
↳204000e+01	2.009750e+01	2.412000e+01		
Daily meals frequency	20000.0	2.864804e+00	6.366486e-01	1.
↳950000e+00	2.020000e+00	3.000000e+00		
Physical exercise	20000.0	4.525030e-01	9.866276e-01	-7.
↳000000e-02	-1.000000e-02	1.000000e-02		
Carbs	20000.0	2.497786e+02	5.510862e+01	1.
↳377200e+02	2.075475e+02	2.429000e+02		
Proteins	20000.0	9.991829e+01	2.204367e+01	5.
↳517000e+01	8.304000e+01	9.714500e+01		
Fats	20000.0	6.661217e+01	1.469928e+01	3.
↳659000e+01	5.534000e+01	6.477000e+01		
Calories	20000.0	2.024418e+03	5.418915e+02	7.
↳810000e+02	1.634000e+03	1.919000e+03		
sugar_g	20000.0	2.453104e+01	1.444610e+01	-6.
↳700000e-01	1.199000e+01	2.406000e+01		

sodium_mg	20000.0	1.258140e+03	7.166190e+02	1.
↪922000e+01	6.451275e+02	1.267650e+03		
cholesterol_mg	20000.0	1.484656e+02	8.738282e+01	-3.
↪890000e+00	7.197000e+01	1.497950e+02		
serving_size_g	20000.0	3.027195e+02	1.157119e+02	9.
↪595000e+01	2.072175e+02	3.000150e+02		
prep_time_min	20000.0	3.351745e+01	1.652687e+01	3.
↪950000e+00	1.858750e+01	3.433000e+01		
cook_time_min	20000.0	6.150216e+01	3.357975e+01	3.
↪350000e+00	3.247750e+01	6.092000e+01		
rating	20000.0	2.989303e+00	1.157692e+00	9.
↪300000e-01	1.940000e+00	3.000000e+00		
Sets	20000.0	4.425059e+00	5.795959e-01	2.
↪960000e+00	4.000000e+00	4.020000e+00		
Reps	20000.0	1.942732e+01	3.679707e+00	4.
↪850000e+00	1.612000e+01	1.990000e+01		
Burns Calories (per 30 min)	20000.0	3.440465e+02	3.213878e+01	1.
↪287500e+02	3.376000e+02	3.486050e+02		
BMI_calc	20000.0	2.492165e+01	6.701144e+00	1.
↪203791e+01	2.009497e+01	2.411910e+01		
cal_from_macros	20000.0	1.998297e+03	4.408484e+02	1.
↪105570e+03	1.661023e+03	1.943130e+03		
pct_carbs	20000.0	4.999830e-01	1.455490e-03	4.
↪924342e-01	4.990538e-01	4.999810e-01		
protein_per_kg	20000.0	1.460142e+00	5.189455e-01	5.
↪167064e-01	1.076294e+00	1.382260e+00		
pct_HRR	20000.0	6.990053e-01	1.448804e-01	3.
↪713439e-01	5.836562e-01	6.862837e-01		
pct_maxHR	20000.0	8.023050e-01	9.661268e-02	5.
↪997889e-01	7.276759e-01	7.948340e-01		
cal_balance	20000.0	7.443087e+02	7.209466e+02	-1.
↪266220e+03	2.614325e+02	6.911900e+02		
lean_mass_kg	20000.0	5.378638e+01	1.249874e+01	3.
↪094626e+01	4.458704e+01	5.120491e+01		
expected_burn	20000.0	8.663523e+02	2.503171e+02	2.
↪198528e+02	7.140983e+02	8.687214e+02		
Burns Calories (per 30 min)_bc	20000.0	8.631802e+19	3.197579e+19	2.
↪491905e+16	6.441978e+19	8.371578e+19		

	75%	max
Age	4.963000e+01	5.967000e+01
Weight (kg)	8.610000e+01	1.307700e+02
Height (m)	1.800000e+00	2.010000e+00
Max_BPM	1.894250e+02	1.996400e+02
Avg_BPM	1.560600e+02	1.698400e+02
Resting_BPM	6.809000e+01	7.450000e+01

Session_Duration (hours)	1.460000e+00	2.020000e+00
Calories_Burned	1.553112e+03	2.890820e+03
Fat_Percentage	2.967603e+01	3.500000e+01
Water_Intake (liters)	3.120000e+00	3.730000e+00
Workout_Frequency (days/week)	4.000000e+00	5.060000e+00
Experience_Level	2.020000e+00	3.050000e+00
BMI	2.856000e+01	5.023000e+01
Daily meals frequency	3.010000e+00	4.040000e+00
Physical exercise	4.000000e-02	4.050000e+00
Carbs	2.839750e+02	4.624900e+02
Proteins	1.136400e+02	1.853400e+02
Fats	7.575250e+01	1.234200e+02
Calories	2.360000e+03	3.641000e+03
sugar_g	3.749000e+01	5.051000e+01
sodium_mg	1.850893e+03	2.527270e+03
cholesterol_mg	2.218400e+02	3.039900e+02
serving_size_g	4.018600e+02	5.075200e+02
prep_time_min	4.794000e+01	6.129000e+01
cook_time_min	8.937500e+01	1.214600e+02
rating	4.000000e+00	5.060000e+00
Sets	5.000000e+00	5.050000e+00
Reps	2.288000e+01	3.012000e+01
Burns Calories (per 30 min)	3.604725e+02	3.817100e+02
BMI_calc	2.856262e+01	5.022954e+01
cal_from_macros	2.271950e+03	3.699540e+03
pct_carbs	5.009103e-01	5.078886e-01
protein_per_kg	1.750495e+00	3.916881e+00
pct_HRR	7.981957e-01	1.073939e+00
pct_maxHR	8.692108e-01	1.047032e+00
cal_balance	1.176290e+03	3.075580e+03
lean_mass_kg	6.193902e+01	9.011737e+01
expected_burn	1.012533e+03	1.477109e+03
Burns Calories (per 30 min)_bc	1.100442e+20	1.756614e+20

1.1.9 Analyse des valeurs manquantes et doublons

```
[7]: missing_values = df.isnull().sum().sort_values(ascending=False)
    duplicates = df.duplicated().sum()

    print("\n--- Valeurs manquantes par colonne (top 10) ---")
    display(missing_values.head(10))

    print(f"\nNombre total de doublons détectés : {duplicates}")
```

```
--- Valeurs manquantes par colonne (top 10) ---
```

```
Age                                0
```



```

Body Part          0
prep_time_min      0
cook_time_min      0
rating             0
Name of Exercise   0
Sets               0
Reps               0
Benefit            0
Burns Calories (per 30 min) 0
dtype: int64

```

Nombre total de doublons détectés : 0

1.1.10 Liste complète des colonnes disponibles

```

[8]: print("\n--- Liste des colonnes du dataset ---")
      print(df.columns.tolist())

```

```

--- Liste des colonnes du dataset ---
['Age', 'Gender', 'Weight (kg)', 'Height (m)', 'Max_BPM', 'Avg_BPM',
'Resting_BPM', 'Session_Duration (hours)', 'Calories_Burned', 'Workout_Type',
'Fat_Percentage', 'Water_Intake (liters)', 'Workout_Frequency (days/week)',
'Experience_Level', 'BMI', 'Daily meals frequency', 'Physical exercise',
'Carbs', 'Proteins', 'Fats', 'Calories', 'meal_name', 'meal_type', 'diet_type',
'sugar_g', 'sodium_mg', 'cholesterol_mg', 'serving_size_g', 'cooking_method',
'prep_time_min', 'cook_time_min', 'rating', 'Name of Exercise', 'Sets', 'Reps',
'Benefit', 'Burns Calories (per 30 min)', 'Target Muscle Group', 'Equipment
Needed', 'Difficulty Level', 'Body Part', 'Type of Muscle', 'Workout',
'BMI_calc', 'cal_from_macros', 'pct_carbs', 'protein_per_kg', 'pct_HRR',
'pct_maxHR', 'cal_balance', 'lean_mass_kg', 'expected_burn', 'Burns Calories
(per 30 min)_bc', 'Burns_Calories_Bin']

```